VGGT-Seg: Visual Geometry Grounded Transformer for Sparse Multi-view 3D Instance Segmentation

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***Abstract*—3D instance segmentation traditionally relies on dense point clouds derived from precise depth maps and camera poses, which significantly hinders the scalability of models in practical deployments due to the high hardware costs and strict calibration requirements for acquiring such precise geometric data. Although recent feed-forward approaches enable scene reconstruction from sparse multi-view images, the resulting geometries often suffer from severe sparsity and background noise, leading to critical sampling biases and supervision misalignments. To address these limitations, we propose VGGT-Seg, an end-to-end 3D instance segmentation framework that operates solely on sparse multi-view image sequences without requiring explicit sensor-provided geometric inputs. Specifically, to mitigate sampling bias, we propose Semantic Prior-driven Query Initialization (SPQI), which integrates global semantic responses via an attention-guided farthest point sampling algorithm to generate a robust hybrid query set. Furthermore, we develop a Topology-rectified Interleaved Decoder (TID) to resolve supervision misalignments and geometric aliasing. By utilizing topological rectification and instance-centric contrastive loss, it explicitly constrains spatial attention to local instance regions. Extensive experiments on standard benchmarks demonstrate that VGGT-Seg achieves highly competitive performance against state-of-the-art geometry-dependent methods.The code is publicly available at https://github.com/EthanW-coder/VGGT-Seg.**

***Index Terms*—3D Instance Segmentation, Sensor Geometry Free, Feed-forward 3D Reconstruction, End-to-end**

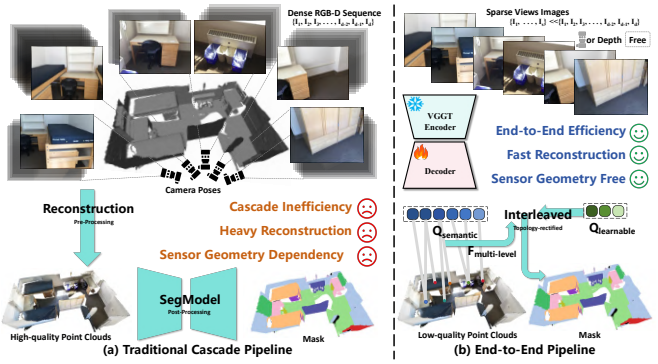


Fig. 1: (a) Traditional cascade pipelines rely on dense RGB-D sequences and camera poses or depth maps for preprocessing to generate high-quality point clouds. The postprocessing stage predicts instance masks via standard backbones and decoders of segmentation models, which is slow and inefficient. (b) We propose VGGT-Seg, a novel end-to-end paradigm. It employs only sparse view images as input, without incorporating any sensor input. Instead of merely utilizing the predictions of Visual Geometry Grounded Transformer (VGGT) [20], our method effectively leverages the internal semantic and geometric priors of VGGT to generate high-quality instance masks in an end-to-end manner.

I. Introduction

3D instance segmentation (3DIS) is a core task in deep learning for 3D domains [1]–[3], with widespread applications in autonomous driving [4], robotic perception [5], and embodied AI [6]. Previous research [7]–[19] typically relies on a preprocessing stage that requires precise calibration of geometric information from multi-view camera poses and depth maps, followed by reconstruction to obtain input point clouds that contain explicit geometric shapes, color, and semantic information. Subsequently, point cloud features are extracted and instance masks are obtained in post-processing. As illustrated in Fig. 1(a), the computational burden and inherent latency of this complex cascade pipeline severely restrict its scalability and efficiency for real-world deployments.

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In practical 3D scenarios (e.g., robotics, mobile devices, and AR glasses), scene geometry and semantic content are often captured solely through multi-view RGB image sequences. Without explicit geometric input, jointly learning 3D structure reconstruction and point-level instance discrimination from images becomes highly difficult. Recent research has demonstrated that feed-forward 3D reconstruction approaches [21]–[24] provide practical solutions for 3D geometry reconstruction from sparse views. An intuitive and straightforward idea involves two stages: reconstructing point clouds from sparse multi-view images, followed by segmenting the point clouds. However, as illustrated in Fig. 2, point clouds obtained from sparse multi-view images exhibit poor quality and geometric deviations compared to the ground truth. The reconstructed geometry can only recover the coarse contours of objects, making it difficult for the initialized queries to cover the complete regions of target instances. Moreover, significant background noise prevents the model from delineating the boundary contours of the targets, which leads to two critical issues: 1) *Sparse geometry-guided sampling bias:* Traditional query sampling methods based on geometric density (e.g., farthest point sampling [25], superpoint [26]) cannot effectively capture the complete spatial distribution of instances. This re-

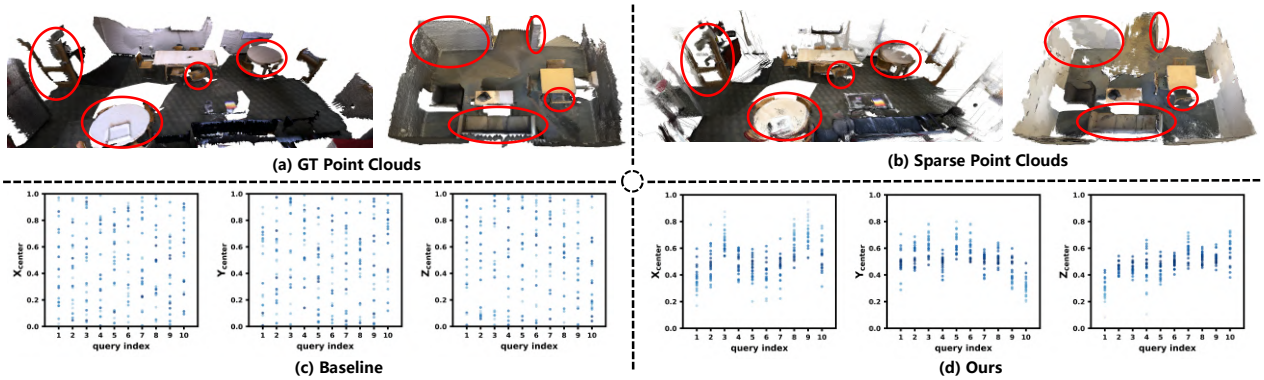


Fig. 2: As an ideal input format, (a) GT point clouds present dense, continuous surfaces with distinctly defined object boundaries. In contrast, (b) sparse point clouds constructed by feed-forward methods [21], [22], [27] exhibit geometric aliasing and degradation in instance representation within space, and are contaminated by background noise, which consequently leads to sampling bias and supervision misalignment. (c) and (d) visualize the spatial coordinates of the matched ground-truth instances for 10 randomly selected queries during the early stages of training. Compared to the (c) baseline, which exhibits severe spatial variance and unstable matching, (d) our method achieves highly concentrated and stable query localization.

sults in initialized queries frequently missing partial instances and suffering from severe localization bias. Consequently, simple geometric criteria fail to locate instance centroids. 2) *Supervision misalignment on sparse points:* Traditional geometric loss functions assume that supervision masks are reliable. However, under coarse point clouds, these sparse points exhibit significant geometric deviations from the ground truth mask locations. Incorrect spatial correspondences lead to unstable gradient signals.

To address the aforementioned challenges, we propose VGGT-Seg, an end-to-end framework for 3DIS that operates solely on sparse multi-view images. As illustrated in Fig. 1(b), VGGT-Seg eliminates the requirement for depth maps and sensor poses. By extensively exploiting the internal geometric and semantic priors of visual foundation models, we effectively bridge the gap between low-level geometric reconstruction and high-level semantic perception. Specifically, we extract multi-level scene features and coarse geometric structures utilizing a frozen visual transformer. To mitigate the sampling bias induced by sparse geometry, we design a **S**emantic **P**rior-driven **Q**uery **I**nitialization **(SPQI)** module, which incorporates global semantic responses into the spatial sampling process via an attention-guided farthest point sampling algorithm, thereby generating a hybrid query set that ensures the robust localization of both explicit and implicit instances. Unlike the supervised query initialization of SGIFormer [28] in dense point clouds, the hybrid queries in SPQI directly extract semantic priors from the shallow attention maps of the frozen visual foundation model DINOv2 [29], thereby avoiding the infeasibility of feature aggregation over voxels or superpoints on sparse coarse geometric structures. Furthermore, to resolve the spatial offset of supervisory signals and the geometric aliasing present in sparse point clouds, we introduce a **T**opology-rectified **I**nterleaved **D**ecoder **(TID)**, which employs a rectification and supervision strategy that introduces stochastic topology queries during the training phase. This step explicitly constrains the spatial attention of queries to local instance regions. Coupled with an instance-centric rectified contrastive loss, this decoding architecture significantly enhances the boundary discrimination capability of the model. To intuitively demonstrate the superiority of our framework in handling sparse inputs, we track the matching behavior of instance queries during the early stages of training. As illustrated in Fig. 2(c), traditional paradigms suffer from severe geometric sampling bias and supervision misalignment on sparse point clouds, leading to chaotic and unstable matching between the queries and the center coordinates $\mathbf{X}_{\mathbf{center}}$, $\mathbf{Y}_{\mathbf{center}}$, $\mathbf{Z}_{\mathbf{center}}$ of ground-truth instances. In contrast, as shown in Fig. 2(d), VGGT-Seg achieves highly stable and concentrated query localization. In summary, the contributions of our work are as follows:

- We propose SPQI, which alleviates the severe localization bias caused by sparse multi-view geometries by incorporating global semantic responses from visual foundation models. Specifically, an attention-guided farthest point sampling strategy is introduced to initialize a robust hybrid query set with improved spatial coverage and semantic reliability.
- We develop TID, which resolves the supervision misalignments and geometric aliasing inherent in coarse point clouds through topology-rectified query rectification and instance-centric rectified contrastive loss.
- Extensive experiments on standard benchmarks demonstrate the superior performance of VGGT-Seg. Without relying on any explicit geometric inputs, the proposed paradigm achieves highly competitive results.

II. Related Work

A. 3D Instance Segmentation

Traditional 3D instance segmentation frameworks are categorized into proposal-based, grouping-based, and Transformer-based architectures. Proposal-based methods [30]–[32] first generate 3D bounding box proposals and subsequently predict instance masks within these localized regions. Grouping-

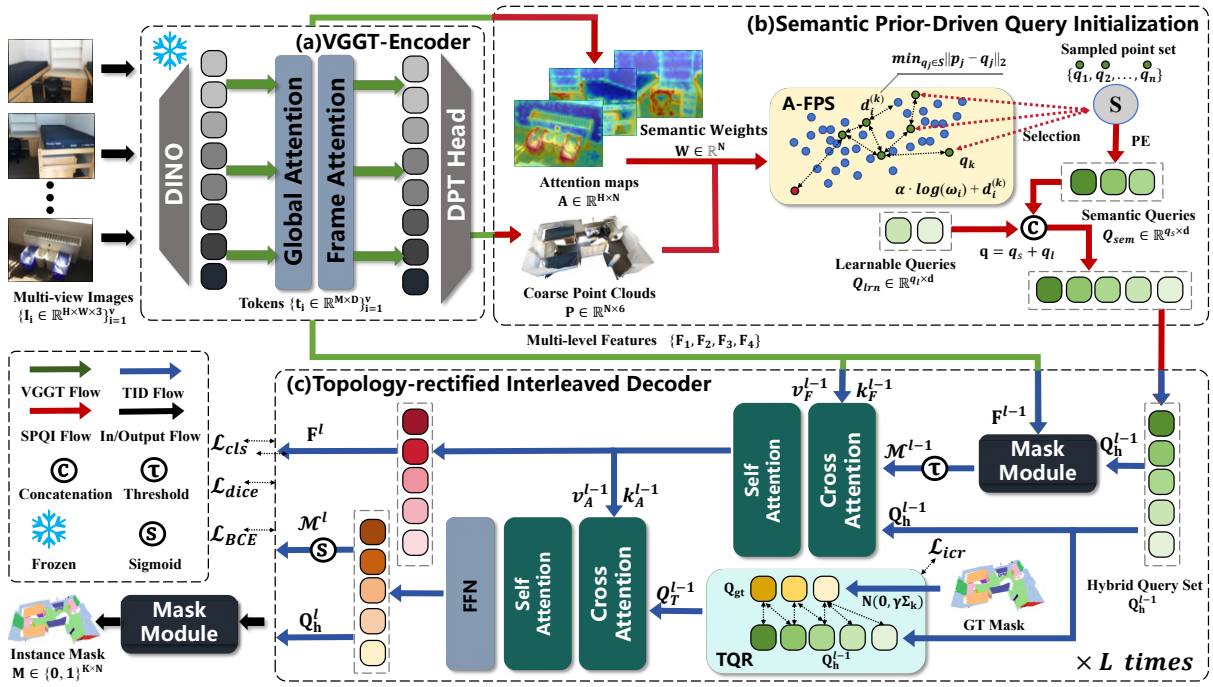


Fig. 3: Overall architecture of the proposed VGGT-Seg. (a) The frozen VGGT-Encoder extracts multi-level scene features and a coarse point cloud. (b) Semantic Prior-driven Query Initialization applies global semantic weights of attention maps to guide Attention-guided Farthest Point Sampling (A-FPS) to generate a hybrid query set, mitigating sampling biases induced by sparse geometry. (c) Topology-rectified Interleaved Decoder alternately embeds geometric perception and semantic discrimination. By utilizing Topology-rectified Query Rectification (TQR) to explicitly constrain spatial attention to local regions, it resolves supervision misalignments and outputs the final 3D instance masks.

based methods [8], [33]–[36] aggregate point-wise semantic predictions and geometric offsets into instance clusters via heuristic algorithms. However, these methods inevitably require time-consuming intermediate clustering steps and rely on manually tuned hyperparameters. In recent years, Transformer-based architectures [9], [10], [28] have significantly boosted performance by pioneering the use of learnable and sampled instance queries, iteratively optimizing a fixed number of instance queries. Subsequent research advancements have further refined this paradigm by introducing auxiliary center regression tasks [37], language-driven semantic priors [12], and progressive query aggregation strategies [28]. Existing paradigms strictly rely on explicit geometric inputs [38], when applied to sparse and noisy structures reconstructed solely from unconstrained multi-view images, they inevitably suffer from severe spatial aliasing and supervision misalignments.

B. Feed-forward 3D Reconstruction

Recently, feed-forward 3D reconstruction has emerged as a promising alternative to traditional optimization-based pipelines. The pioneering work, DUSt3R [21], formulates multi-view 3D reconstruction as a dense pointmap regression task directly from unconstrained image sets, successfully eliminating the necessity for complex camera calibration. Building upon this foundation, subsequent studies have significantly improved its scalability and efficiency. Specifically, MASt3R [22] enhances dense 3D correspondences and image matching, whereas Fast3R [39] and WinT3R [24] introduce highly efficient architectures capable of processing large-scale image collections and online streaming data in a single forward pass, respectively. Furthermore, recent architectures such as the VGGT [20] and its accelerated derivative, FastVGGT [40], demonstrate that strong 3D geometric priors can be effectively inferred directly from 2D foundation models [41]–[45] without explicit spatial inputs.

Driven by these geometric advancements, researchers have increasingly extended feed-forward reconstruction paradigms to various downstream 3D perception tasks. Notable explorations include open-vocabulary scene segmentation [46], simultaneous geometric reconstruction and semantic understanding [47], [48]. In recent studies, VGGT-Det [27] was the first to achieve object detection in coarse point clouds, providing a reference for the paradigm of extracting priors within VGGT. However, unlike 3D bounding box regression, directly transforming sparsely reconstructed geometries into point-level instance discrimination remains challenging. The concurrent work SegVGGT [49] further explores this gap by embedding instance queries into VGGT to interact with 2D image features, generating multi-view 2D masks that are then back-projected into 3D space using predicted depths. In contrast, starting from the fundamental challenges of sparse geometry, we employ a semantic-guided hybrid query initialization to address sampling bias and propose a topology-rectified regularization technique to mitigate supervision misalignment, thereby achieving end-to-end point-level instance mask prediction directly in 3D space.

III. METHODOLOGY

A. Problem Statement

Given a set of sparse multi-view RGB images $\{\mathbf{I}_i\}_{i=1}^{v} = \{I_1, \ldots, I_v\}$, and in the absence of any sensor-provided multi-view camera poses or depth maps, the objective is to implicitly reconstruct the sparse 3D geometric structure of the scene from $\{\mathbf{I}_i\}_{i=1}^{v}$ and partition the structure into K distinct instances. Formally, this task aims to train an end-to-end learnable network f_θ that directly maps the image set $\{\mathbf{I}_i\}_{i=1}^{v}$ to 3D instance segmentation results, which can be expressed as:

$$f_\theta(\{\mathbf{I}_i\}_{i=1}^{v}) = \{\mathbf{P}, \mathbf{M}, \mathbf{Y}\} \tag{1}$$

where $\mathbf{P} \in \mathbb{R}^{N\times 6}$ denotes the inferred coarse point cloud, N is the number of points, and each point comprises (x, y, z, r, g, b). $\mathbf{M} \in \{0,1\}^{K\times N}$ denotes a set of binary instance masks, and $\mathbf{Y} \in \mathbb{R}^{K\times C}$ denotes a set of semantic labels, with C being the number of predefined semantic categories.

B. Semantic Prior-driven Query Initialization

In point clouds reconstructed from sparse multi-view images, significant geometric aliasing arises between background noise and instance boundaries. This degradation causes traditional query initialization methods [7], [9], [50] to fail, resulting in severe localization bias and preventing model convergence. To overcome the sampling bias induced by sparse geometry guidance, we propose **S**emantic **P**rior-driven **Q**uery **I**nitialization **(SPQI)**. As illustrated in Fig. 3(b), it aims to explicitly intervene in the 3D spatial sampling process by introducing rich global semantics from 2D visual foundation models, and further enhances the diversity of the query set and convergence speed through a hybrid query approach.

Geometric-semantic Prior Extraction: We follow prior work [20] to build an encoder-decoder transformer for extracting geometric priors, with all parameters kept frozen, thereby ensuring the stability of the geometric priors without needing to relearn 3D geometry from sparse images, as illustrated in Fig. 3(a). Specifically, the input multi-view images $\{\mathbf{I}_i\}_{i=1}^{v}$ are patchified into token representations $\{t_i \in \mathbb{R}^{M\times D}\}_{i=1}^{v} = \{t_1, \ldots, t_v\}$ via frozen DINOv2 [29], with M tokens per view and D as the token dimension. Subsequently, the model alternates between frame-wise and global attention mechanisms to form geometry-aware token representations. Specifically, the frame-wise attention operates independently within the tokens of each individual frame to preserve the viewpoint-specific spatial layout, whereas the global attention jointly processes tokens across all frames to establish multi-view consistency and implicit epipolar geometry. Finally, these multi-level scene features $\mathbf{F}$ are fed into a reconstruction decoder to predict continuous depth maps and camera poses. They are back-projected to obtain coarse point cloud $\mathbf{P} = \{p_1, p_2, \ldots, p_N\}$. This process can be represented as:

$$\{\mathbf{P}, \mathbf{A}, \mathbf{F}\} = \textbf{VGGTEncoder}(\{\mathbf{I}_i\}_{i=1}^{v}) \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{H\times N}$ denotes attention maps. Notably, $\mathbf{A}$ is extracted from the first layer of the frozen DINOv2. This is because while deeper Transformer layers capture highly abstract semantic representations, they suffer from severe spatial resolution degradation, leading to ambiguous instance boundaries. In contrast, shallow network layers effectively preserve fine-grained spatial details and sharp instance contours. It is noteworthy that VGGT itself does not require pre-training on semantic tasks. Because the DINOv2 backbone inherently retains rich semantic representations, it enables us to directly leverage powerful 2D semantic priors to guide sparse 3D geometry.

Algorithm 1 Attention-guided Farthest Point Sampling

Require: coarse point cloud $\mathbf{P} = \{p_1, p_2, \ldots, p_N\}$ Semantic weights $\mathbf{W} = \{w_1, w_2, \ldots, w_N\}$ Target query count n, Trade-off parameter α
Ensure: Sampled query point set $\mathbf{S} = \{q_1, ..., q_n\}$

$\mathbf{S} \leftarrow \emptyset$
/* Select first point maximizing semantic weight */
$q_1 \leftarrow \arg\max_{p_i \in P}(w_i)$
$\mathbf{S} \leftarrow \mathbf{S} \cup \{q_1\}$
for $k = 2$ **to** n **do**
  for each $p_i \in \mathbf{P} \setminus \mathbf{S}$ **do**
    $d_i^{(k)} \leftarrow \min_{q_j \in \mathbf{S}} ||p_i - q_j||_2$
  end for
  /* Select the point maximizing the joint energy */
  $q_k \leftarrow \arg\max_{p_i \in \mathbf{P}\setminus\mathbf{S}} \left(\alpha \cdot \log(w_i) + d_i^{(k)}\right)$
  $\mathbf{S} \leftarrow \mathbf{S} \cup \{q_k\}$
end for
return $\mathbf{S}$

Attention-guided Farthest Point Sampling: After obtaining the features containing semantic responses, we assign a semantic prior weight $\mathbf{W} = \{w_1, w_2, \ldots, w_N\} \in \mathbb{R}^N$ to each 3D point in the coarse point cloud $\mathbf{P} \in \mathbb{R}^{N\times 6}$, where $w_i \in \mathbf{W}$ denotes the confidence score of the i-th 3D point p_i belonging to a foreground instance. Unlike the heuristic-based linear weighting approach adopted in previous work [27], we first aggregate the multi-head attention scores by averaging across the H heads, and then apply a spatial Softmax over all N points to obtain a valid confidence distribution. The weight w_i for the i-th point is formulated as:

$$w_i = \frac{\exp(s_i)}{\sum_{j=1}^{N}\exp(s_j)}, \quad \text{where} \quad s_i = \frac{1}{H}\sum_{h=1}^{H} A_{h,i} \tag{3}$$

where $A_{h,i}$ denotes the attention score of the h-th attention head for the i-th point. Crucially, the correspondence between the attention map $\mathbf{A} \in \mathbb{R}^{H\times N}$ and the N 3D points is established through a deterministic spatial back-projection. Utilizing the camera parameters and depth predicted by VGGT-encoder, both reside in the exact same reference coordinate system, thereby guaranteeing a strict one-to-one spatial mapping. Traditional FPS [25] relies solely on the Euclidean geometric distance between points. This renders them prone to generating invalid queries in sparse regions corrupted by substantial background noise. To address this, we propose a novel **A**ttention-guided **FPS (A-FPS)** algorithm, as outlined in Alg. 1. During the initialization phase, we select the point with the highest global semantic weight as

the initial anchor $q_1 = \arg\max_{p_i \in P}(w_i)$. This ensures that the sampling process consistently originates from the most confident foreground center. In the subsequent iterative sampling steps (i.e., the k-th step), for each unselected candidate point p_i, we first compute its minimum Euclidean distance $d_i^{(k)}$ to the currently selected point set $\mathbf{S}$. To rectify the spatial bias introduced by purely geometric sampling, we formulate a joint energy function that couples the geometric distribution of the point cloud with its corresponding semantic response. The next optimal query point is selected by maximizing this joint energy function, which is expressed as:

$$q_k = \arg\max_{p_i \in \mathbf{P} \setminus \mathbf{S}} \left(\alpha \cdot \log(w_i) + d_i^{(k)} \right) \tag{4}$$

where α is a trade-off hyperparameter balancing the strength of the semantic guidance and the uniformity of the spatial distribution. It is important to note that the selected set $\mathbf{S}$ is explicitly anchored by the semantic maximum (i.e., q_1), and therefore $d_i^{(k)}$ inherently acts as a semantic-conditioned spatial metric. To mathematically bridge the scale disparity between the semantic confidence w_i and the Euclidean distance $d_i^{(k)}$, the logarithmic term $\log(w_i)$ maps the normalized probability $(0, 1)$ into an energy penalty ranging from $(-\infty, 0)$, effectively smoothing extreme peaks in the semantic distribution, thereby preventing the sampled points from over-concentrating in a single high-confidence region. A-FPS efficiently generates n high-quality query points that exhibit both broad spatial coverage and high semantic responses. It maximizes spatial exploration while compelling the queries to converge toward highly semantic instance centers. We employ sinusoidal positional encoding to map the sampled query point set into a high-dimensional feature space, yielding the semantic queries $Q_{sem} \in \mathbb{R}^{q_s \times d}$.

Hybrid Query Initialization: Previous studies [27] have shown that semantic queries are highly robust to surface sparsity in bounding box regression. In contrast, point-level mask inference is more sensitive to geometric aliasing. Although the semantic queries Q_{sem} provided by A-FPS can capture salient explicit targets in the scene, to recover instances with weaker visual saliency, we introduce learnable parameterized queries to complement the sampled queries $Q_{lrn} \in \mathbb{R}^{q_l \times d}$, adaptively capturing implicit instances within the global scene that might be overlooked by the semantic guidance. Q_{sem} and Q_{lrn} complement each other, with the former providing explicit global priors and the latter acting as an implicit local supplement. We concatenate the semantic-guided queries with the learnable parameterized queries to form a hybrid query set $Q_h \in \mathbb{R}^{q \times d}$:

$$Q_h = \text{Concat}[Q_{sem}; Q_{lrn}] \in \mathbb{R}^{(q_s+q_l) \times d} \tag{5}$$

Hybrid query initialization set Q_h ensures the rapid localization of salient instances, serving as a powerful initial prior input for the subsequent Interleaved Denoising Decoder.

C. Topology-rectified Interleaved Decoder

Due to the geometric aliasing in coarse point clouds, traditional global attention mechanisms tend to introduce extensive redundant background features within highly sparse and noisy geometric fields, leading to query degradation. We design a **T**opology-rectified **I**nterleaved **D**ecoder **(TID)**, which employs a dual-stream progressive optimization strategy and explicitly imposes spatial constraints, alternately embedding low-level geometric perception and high-level semantic discrimination into the query updating process, as illustrated in Fig. 3(c). Specifically, given the hybrid query set Q_h^{l-1} and the scene features $\mathbf{F}^{l-1}$, TID evaluates the confidence that each spatial point belongs to a specific instance by computing their dot product similarity. This generates a discrete attention mask $\mathcal{M}^{l-1}$ for the current layer, formulated as:

$$\mathcal{M}_{i,j}^{l-1} = \begin{cases} 0, & \text{if } \sigma\left(\mathbf{F}^{l-1} \cdot (Q_h^{l-1})^T\right)_{i,j} > \tau \\ -\infty, & \text{otherwise} \end{cases} \tag{6}$$

where σ denotes the Sigmoid activation function, and τ is a predefined threshold for binarization. Following the standard masked attention convention, we assign 0 to positions where the sigmoid score exceeds τ to permit attention, and $-\infty$ otherwise to strictly block attention before the Softmax operation. This ensures that each query attends exclusively to its predicted local instance region. The multi-scale geometric and semantic features extracted from VGGT are linearly projected into keys k_F^{l-1} and values v_F^{l-1}. Using the hybrid queries Q_h^{l-1} as the query vectors, the mask $\mathcal{M}^{l-1}$ imposes an explicit spatial constraint on the cross-attention mechanism. This forces each query to strictly attend to its predicted local instance region rather than the global scene:

$$\hat{Q}_h^{l-1} = \text{Softmax}\left(\frac{Q_h^{l-1}(k_F^{l-1})^T}{\sqrt{d}} + \mathcal{M}^{l-1} \right) v_F^{l-1} \tag{7}$$

Subsequently, the updated query $\hat{Q}_h^{l-1}$ are fed into the Self-Attention to suppress redundant predictions caused by multiple similar queries focusing on the same instance. We then obtain the updated scene features $\mathbf{F}^l$.

Topology-rectified Query Rectification: To alleviate the spatial offset problem of supervisory signals on sparse point clouds, we introduce **T**opology-rectified **Q**uery **R**ectification **(TQR)** in decoder layer. We extract prior centers with geometric representation capability from the ground truth (GT) labels of the dataset. Given the ground truth 3D point set mask $M_k \in \mathbb{R}^{N_k \times 3}$ of the k-th instance, we compute its geometric centroid c_k and the covariance matrix Σ_k of its spatial distribution. Next, introduce adaptive perturbations to each centroid that conform to the topological distribution of the instance, generating stochastic topology anchors c'_k:

$$c'_k = c_k + \delta_k, \quad \delta_k \sim \mathcal{N}(0, \gamma\Sigma_k) \tag{8}$$

where $\mathcal{N}$ denotes the Gaussian distribution, and γ is a scaling factor controlling the perturbation strength. Subsequently, for the K ground-truth instances in the current scene, we extract their centers, apply perturbations, and map them into a high-dimensional feature space via sinusoidal position encoding, thereby generating the topology query $Q_{gt} \in \mathbb{R}^{K \times d}$. We aggregate the hybrid query features $Q_h^{l-1} \in \mathbb{R}^{q \times d}$ with the embedded strong prior topology information Q_{gt} via

concatenation, thereby obtaining the modulated latent queries $Q_T^{l-1} \in \mathbb{R}^{(q+K)\times d}$ infused with topological noise priors:

$$Q_T^{l-1} = \mathbf{W}_2 \cdot \phi(\mathbf{W}_1 \cdot \text{Concat}[Q_h^{l-1}; Q_{gt}])^T \tag{9}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ represent the learnable parameters of the linear layers, which act strictly along the feature dimension d as point-wise transformations, and ϕ is the GELU function. In the process of progressive query updating, TQR serves to explicitly constrain spatial attention to local instance regions. Then, we employ the Hungarian algorithm [51] to compute the minimum matching cost between predicted masks and ground truth masks, yielding the optimal matching permutation π. The modulated query $\tilde{q}_{\pi(k)}$ achieving the minimum matching cost is regarded as the positive sample for the k-th instance, while queries failing to match any instance are designated as negative samples to be suppressed. In the contrastive loss, we introduce a dynamic temperature weight $\omega_{j,k}$ based on Euclidean distance to suppress the interference of hard negative samples near the instance center with mask boundaries. The formula is as follows:

$$\omega_{j,k} = \exp\left(-\frac{\|p_j - c_k\|_2^2}{2\rho^2}\right) \tag{10}$$

where p_j denotes the decoded 3D spatial coordinate of the j-th negative query. The instance-centric rectified contrastive loss $\mathcal{L}_{\text{icr}}$ employs q^{gt} as an anchor to attract the matched positive sample $\tilde{q}_{\pi(k)}$, while simultaneously leveraging $\omega_{j,k}$ to forcefully repel spatially confusing negative samples $\tilde{q}_j$:

$$S(x, y) = \exp\left(\text{sim}(x, y)/\epsilon\right), \tag{11}$$

$$\mathcal{L}_{\text{icr}} = -\frac{1}{K}\sum_{k=1}^{K} \log \frac{S(\tilde{q}_{\pi(k)}, q^{\text{gt}})}{S(\tilde{q}_{\pi(k)}, q^{\text{gt}}) + \sum_{\tilde{q}_j \in \mathcal{N}} \omega_{j,k} \cdot S(\tilde{q}_j, q^{\text{gt}})} \tag{12}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity and ϵ represents the temperature scaling coefficient in contrastive learning. During the inference phase, topology query Q_{gt} is not generated, and TQR degenerates into a decoding process without external topological supervision. The source for updating modulated latent queries Q_T^{l-1} relies exclusively on SPQI.

Prediction Head: We leverage global view semantics to further optimize the boundary perception capabilities of the queries. Similarly, the modulated latent queries Q_T^{l-1}, alongside the keys k_A^{l-1} and values v_A^{l-1} of updated scene features $\mathbf{F}^l$, are fed into cross-attention and self-attention mechanisms. Subsequently, they are passed into a Feed-Forward Network (FFN), where a GELU activation function performs non-linear transformations within the high-dimensional feature space, outputting the updated instance queries Q_h^l for the current layer. The aforementioned decoding process is repeated L times, fully decoupling the topological and geometric information in the scene features from the appearance and semantic information in the queries. At the end layer of the decoder, by utilizing a Multi-Layer Perceptron (MLP) and computing the dot product similarity between the updated hybrid queries Q_h^l and the scene features F^l, binary instance mask $\mathbf{M} \in \{0,1\}^{K\times N}$ is generated:

$$\mathbf{M} = \sigma\left((\mathbf{F}^l) \cdot \text{MLP}(Q_h^l)^T\right) \tag{13}$$

where σ denotes the Sigmoid function. Simultaneously, to determine the semantic attributes of each instance, the updated hybrid queries Q_h^l are fed into a shared MLP to predict the corresponding semantic label $\mathbf{Y} \in \mathbb{R}^{K\times C}$.

D. Bipartite Matching and Loss Function

Since there is no inherent sequential order between the set of predicted instances and the set of ground-truth labels, we treat the pairing task between them as an optimal bipartite matching problem [57]. Formally, let i denote the prediction proposal and j denote the ground-truth label, the pairwise matching cost matrix $\mathcal{C}_{i,j}$ is formulated as follows:

$$\mathcal{C}_{i,j} = -\lambda_{cls} p_{i,c_j} + \lambda_{mask}\mathcal{C}_{i,j}^{mask} \tag{14}$$

where p_{i,c_j} indicates the probability for the category c_j of i-th proposal. We combine the binary cross-entropy loss and the Dice loss [58] to form mask matching cost $\mathcal{C}_{i,j}^{mask}$:

$$\mathcal{C}_{i,j}^{mask} = \lambda_{BCE}\mathcal{L}_{BCE}(\mathbf{M}_i, \hat{\mathbf{M}}_j) + \lambda_{dice}\mathcal{L}_{dice}(\mathbf{M}_i, \hat{\mathbf{M}}_j) \tag{15}$$

where $\mathbf{M}_i$ and $\hat{\mathbf{M}}_j$ represent the predicted instance mask and the corresponding ground truth mask, respectively. We utilize the Hungarian algorithm [51] to efficiently solve for the optimal permutation $\hat{c}$, which minimizes the assignment cost:

$$\hat{c} = \arg\min_{c} \sum_{i=1}^{N} \mathcal{C}_{i,c(i)} \tag{16}$$

where $c(i)$ denotes the index of the prediction assigned to the i-th ground truth under permutation c. After establishing the optimal matching relationship, we adopt the cross-entropy loss to supervise the classification $\mathcal{L}_{cls}$. The mask loss $\mathcal{L}_{mask}$ is calculated similarly to $\mathcal{C}_{i,j}^{mask}$. We compute a weighted sum of the aforementioned loss terms to obtain the total loss function $\mathcal{L}_{total}$ for end-to-end network optimization:

$$\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{mask}\mathcal{L}_{mask} + \lambda_{icr}\mathcal{L}_{icr} \tag{17}$$

where $\lambda_{cls}, \lambda_{mask}, \lambda_{icr}$ are hyperparameters used to balance the distinct loss components.

IV. Experiments

A. Implementation Details

During the training process, the parameters of the VGGT-encoder [20] are kept frozen. We employ a basic feature aggregation method to sequentially extract multi-level encoded features from the 4th, 11th, 17th, and 23rd layers of the encoder. The network is trained for 120 epochs using the AdamW [59] optimizer with a weight decay of 0.05. In each training epoch, we randomly sample between 12 and 48 frames per scene as input sequences. During inference, the model takes only sparse RGB images as input, without requiring camera poses or depth maps, and we dynamically sample input frames per scene with adaptive strides, up to a maximum of 80 frames. Following previous work [27], the base learning rate for the newly initialized parameters in VGGT-Seg is set to 3×10^{-4}. To ensure stable model convergence, we apply a cosine annealing schedule, which smoothly decays

TABLE I: Comparison of VGGT-Seg with different methods on the validation set of ScanNet200 [52]. The best results are shown in **bold**. †denotes the cascaded pipeline where multi-view images are pre-reconstructed by VGGT [20].

Info.	Method	All			Head		Common		Tail	
		mAP	mAP_{50}	mAP_{25}	mAP	mAP_{50}	mAP	mAP_{50}	mAP	mAP_{50}
Point Cloud	Mask3D [9]	27.4	37.0	42.3	40.3	55.0	22.4	30.6	18.2	23.2
	SPFormer [53]	25.2	33.8	39.6	–	–	–	–	–	–
	QueryFormer [10]	28.1	37.1	43.4	–	–	–	–	–	–
	MAFT [37]	29.2	38.2	43.3	–	–	–	–	–	–
	OpenMask3D [54]	15.4	19.9	23.1	17.1	–	14.1	–	14.9	–
	Open3DIS [55]	18.6	23.1	27.3	24.7	–	16.9	–	13.3	–
	Open-YOLO3D [56]	24.7	31.7	36.2	27.8	–	24.3	–	**21.6**	–
	SGIFormer [28]	29.2	39.4	44.2	–	–	–	–	–	–
	OneFormer3D [7]	**30.2**	**40.9**	**44.6**	**42.0**	**57.7**	**27.0**	**36.3**	20.1	**26.6**
RGB-D	Mask3D [9]	15.5	21.4	24.3	21.9	31.4	13.0	17.2	7.9	10.3
	ODIN-ResNet50 [38]	25.6	36.9	43.8	34.8	51.1	23.4	33.4	17.8	24.9
	ODIN-Swin-B [38]	**31.5**	**45.3**	**53.1**	**37.5**	**54.2**	**31.6**	**43.9**	**24.1**	**36.6**
Multi-view	Mask3D† [9]	5.9	7.4	10.1	8.5	11.2	4.8	6.1	3.2	4.1
	OneFormer3D† [7]	8.2	10.7	13.7	12.5	16.2	7.4	9.8	3.9	5.3
	VGGT-Seg (Vanilla)	18.3	20.5	28.9	17.4	27.5	17.9	19.0	9.5	12.4
	VGGT-Seg (Full)	**31.2**	**42.7**	**52.7**	**39.5**	**52.2**	**29.8**	**42.5**	**23.1**	**34.3**

the learning rate to a minimum of 1×10^{-4}. All experiments are conducted on 4 NVIDIA RTX 4090 GPUs with a batch size of 2 per GPU. For the SPQI, we configure the model with 200 semantic queries Q_{sem} and 100 learnable queries Q_{lrn}. TID consists of $L = 6$ stacked layers. The balance weights for the loss functions are set as $\lambda_{cls} = \lambda_{icr} = 0.5, \lambda_{mask} = 1.0$. The mask binarization threshold τ in Eq. (6) is set to 0.5, topological perturbation scaling factor γ in Eq. (8) is set to 0.15. Additionally, the ground-truth depths and poses provided by the dataset are used solely for point-to-point mask supervision signals and metric computation during validation. For a fair comparison under the sparse multi-view setting, we construct cascaded baselines based on Mask3D [9], ISBNet [8], Q-Former [10], and OneFormer3D [7], indicated by the † symbol in the comparison tables. Specifically, sparse RGB images are first processed by VGGT to reconstruct coarse 3D point clouds, which are subsequently used as inputs to the corresponding segmentation models for retraining, with training hyperparameters kept consistent with those in the original baseline papers. For coordinate alignment and evaluation methodology, please refer to Supplementary Material A, for hyperparameter ablation studies, please refer to Supplementary Material C.

TABLE II: Comparison of VGGT-Seg with different methods on the validation set of ScanNetv2 [60]. The best results are shown in **bold**.

Info.	Method	mAP	AP_{50}	AP_{25}
Point Cloud	PointGroup [33]	34.8	56.9	71.3
	HAIS [34]	43.5	64.4	74.6
	SoftGroup [36]	45.8	67.6	78.9
	ISBNet [8]	56.8	73.3	81.3
	DyCo3D [35]	35.4	57.6	72.9
	Mask3D [9]	55.2	73.7	82.9
	SPFormer [53]	56.3	73.9	82.9
	Q-Former [10]	**56.5**	**74.2**	**83.3**
RGB-D	Mask3D [9]	43.9	60.0	69.9
	ODIN-ResNet50 [38]	47.8	69.8	**83.6**
	ODIN-Swin-B [38]	**50.0**	**71.0**	**83.6**
Multi-view	ISBNet† [8]	13.8	17.3	20.5
	Mask3D† [9]	9.7	14.7	19.6
	Q-Former† [10]	15.1	20.4	28.7
	OneFormer3D† [7]	12.5	16.9	21.1
	VGGT-Seg (Vanilla)	21.8	37.1	49.8
	VGGT-Seg (Full)	**48.9**	**70.1**	**81.7**

B. Main Results

To comprehensively demonstrate our strong capability in multi-view sparse 3D instance segmentation, we explicitly categorize the methods evaluated in Tables I and II into three distinct groups based on their inference-time input modalities: (a) Point Cloud-based methods [7]–[10], [28], [33]–[37], [53]–[56], which rely on geometric inputs from offline mesh reconstruction. (b) RGB-D-based methods [9], [38], which utilize high-quality geometric information acquired by depth sensors. (c) Multi-view-based methods [7]–[10], which operate solely on sparse RGB images without any geometric sensor assistance and constitute the same-modality comparison group for our VGGT-Seg. To rigorously isolate the performance gains contributed by our proposed modules from the inherent advantages of end-to-end training with frozen foundation models, we introduce a VGGT-Seg (Vanilla), which employs the frozen VGGT encoder combined with standard FPS and a traditional mask decoder [38], entirely omitting the proposed SPQI and TID modules.

ScanNet200: As shown in Tab. I, we comprehensively evaluate our method on the ScanNet200 [52] validation set. Within the category of multi-view-based methods, the vanilla

TABLE III: Comparison of VGGT-Seg with RGB-D based method on the validation set of AI2THOR 3D [61] and Matterport3D [62]. The best results are shown in **bold**.

Info.	Method	AI2THOR 3D			Matterport3D (21class)		Matterport3D (160class)	
		mAP	mAP_{50}	mAP_{25}	mAP	mAP_{50}	mAP	mAP_{50}
Point Cloud	Mask3D [9]	60.6	70.8	76.6	22.9	55.9	11.3	23.9
RGB-D	Mask3D [9]	–	–	–	7.2	16.8	2.5	10.9
	ODIN-ResNet50 [38]	63.8	**73.8**	**80.2**	22.5	56.4	11.5	27.6
	ODIN-Swin-B [38]	**64.3**	73.7	78.6	**24.7**	**63.8**	**14.5**	**36.8**
Multi-view	VGGT-Seg (Vanilla)	29.5	37.1	40.4	9.6	17.3	4.1	12.5
	VGGT-Seg (Full)	**64.2**	**73.4**	**78.9**	**24.1**	**61.0**	**13.8**	**34.9**



Fig. 4: Qualitative comparison analysis on the ScanNetv2 validation [60]. This study presents the ground truth (GT) instance annotations, the baseline [10], and VGGT-seg. The comparison results are highlighted with red circles, and different colors are used in each subfigure to distinguish different instances. Please zoom in for the best visualization results.

baseline VGGT-Seg (Vanilla) achieves an overall mAP of 18.3, outperforming the best cascaded competitor OneFormer3D† (8.2 mAP) by a significant margin of 10.1 mAP. This confirms the inherent benefit of the unified end-to-end representation over decoupled two-stage pipelines. More importantly, the proposed VGGT-Seg (Full) demonstrates a decisive advantage over both cascaded baseline [10] and existing end-to-end frameworks. Traditional state-of-the-art models exhibit severe performance degradation when transitioning from ground-truth dense point clouds to sparse geometries reconstructed solely from multi-view images. For instance, while OneFormer3D [7] achieves an impressive overall **mAP** of 30.2 given ideal point cloud inputs, its performance plummets to a mere 8.2 **mAP** under a cascaded multi-view setting. Standard geometry-dependent sampling and supervision mechanisms fail drastically on sparse and noisy point clouds, underscoring the fundamental limitations of decoupled reconstruction and segmentation pipelines. Achieving an overall **mAP** of 31.2, VGGT-Seg completely surpasses the cascaded OneFormer3D [7] by a substantial margin of 23.0. This significant improvement demonstrates that implicitly leveraging visual foundation models for joint geometric and semantic optimization effectively bridges the representation gap caused by the absence of explicit 3D sensors. Remarkably, despite relying strictly on sparse multi-view images without any depth maps or camera parameters, VGGT-Seg matches or even exceeds the performance of architectures equipped with explicit geometric inputs. It outperforms the point cloud-based method [9], [10], [37], [53], OneFormer3D requires high-quality point clouds by 1.0 in overall **mAP**. Furthermore, compared to RGB-D-based methods that simultaneously utilize images, precise depth maps, and camera parameters, VGGT-Seg achieves highly competitive results. Its overall **mAP** is merely 0.3 lower than that of ODIN-Swin-B [38], while entirely eliminating the reliance on hardware-level geometric measurements. It is worth noting that the ScanNet200 [52] dataset exhibits a high degree of class imbalance regarding rare categories. While cascaded methods nearly fail on the Tail subset, with their **mAP** scores dropping to single digits, VGGT-Seg maintains a stable **mAP** of 23.1. This indicates that the semantic priors directly extracted from 2D foundation models successfully compensate for geometric sparsity, providing highly discriminative features for challenging and infrequent instances.

ScanNetv2: As shown in Tab. II, despite the complete absence of sensor-provided depth maps or precise point clouds, VGGT-Seg demonstrates remarkable competitiveness on the ScanNetv2 [60] validation set when compared to geometry-dependent architectures. Our approach achieves an overall **mAP** of 48.9 and an $\mathbf{mAP}_{50}$ of 70.1, successfully outperforming several established point cloud-based methods including PointGroup [33], HAIS [34], and SoftGroup [36], which rely on ideal geometric inputs. Furthermore, within the RGB-D category, VGGT-Seg surpasses ODIN-ResNet50 [38] by 1.1 **mAP** and exceeds the RGB-D variant of Mask3D by

TABLE IV: Ablation study of VGGT-Seg components on the ScanNet val [60] dataset. We analyze the impact of the proposed components, specifically the **S**emantic **P**rior-driven **Q**uery **I**nitialization **(SPQI)** and the **T**opology-rectified **I**nterleaved **D**ecoder **(TID)** on the overall segmentation performance. ±represent performance change relative to the vanilla or full model.

Exp.	Settings	SPQI			TID		Metrics		
		A-FPS	Q_{sem}	Q_{lrn}	TQR	$\mathcal{L}_{icr}$	mAP	AP_{50}	AP_{25}
1	baseline						15.1	20.4	28.7
2	VGGT-Seg (Vanilla)						21.8	37.1	49.8
3	w/ semantic priors	✓	✓				29.8+8.0	47.6+10.5	61.0+11.2
4	w/ hybrid query		✓	✓			26.2+4.4	39.8+2.7	53.9+4.1
5	w/ TQR				✓		28.3+6.5	47.2+10.1	59.3+9.5
6	w/ TID				✓	✓	32.9+11.1	52.1+15.0	67.6+17.8
7	VGGT-Seg (Full)	✓	✓	✓	✓	✓	48.9	70.1	81.7
8	w/o A-FPS		✓	✓	✓	✓	42.7-6.2	62.4-7.7	73.1-8.6
9	w/o $\mathcal{L}_{icr}$	✓	✓	✓	✓		46.6-2.3	66.9-3.2	79.0-2.7
10	w/o TID	✓	✓	✓			38.4-10.5	56.5-13.6	70.7-11.0

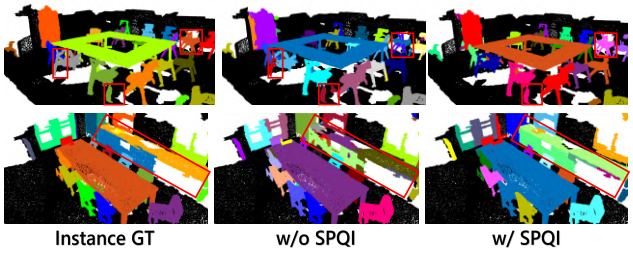


Fig. 5: Ablation study of SPQI on the ScanNet val [60] dataset. Red boxes highlight semantic omissions and instance misidentifications occurring in the absence of SPQI.

a significant margin of 5.0 **mAP**. Although a marginal gap remains when compared to the absolute multi-modal state-of-the-art model ODIN-Swin-B [38], achieving such comparable results using only unconstrained images confirms that the extracted semantic priors effectively compensate for the lack of dense spatial measurements. As shown in Fig. 4, compared to the baseline, VGGT-Seg generates extremely clear instance boundaries, with fewer instance omissions and errors.

AI2THOR 3D and Matterport3D: To further validate the generalization capability of VGGT-Seg across different scene distributions, we conduct extended experiments on the AI2THOR 3D [61] and Matterport3D [62] benchmarks, with results presented in Tab. III. On the AI2THOR 3D dataset, which is collected from simulation environments, VGGT-Seg achieves an overall accuracy of 64.2 **mAP**, nearly on par with ODIN-Swin-B [38] (64.3 **mAP**), the best-performing RGB-D method. Notably, VGGT-Seg attains segmentation performance comparable to that of the optimal multimodal approach during inference without any explicit geometric input, fully demonstrating its capability to reconstruct semantic geometry from sparse multi-view images. On the more challenging Matterport3D dataset, this advantage becomes even more pronounced. We conduct evaluations under both the standard 21-class protocol and the more challenging 160-class protocol. VGGT-Seg achieves 24.1 **mAP** and 61.0 **mAP**$_{50}$ on the 21-class split, trailing ODIN-Swin-B [38] by only 0.6 **mAP**; on the 160-class split, VGGT-Seg attains 13.8 **mAP** and 34.9 **mAP**$_{50}$, remaining on par with the best RGB-D methods.

C. Ablation Study

Impact of each component: Tab. IV comprehensively evaluates the impact of each proposed component in VGGT-Seg. The transition from the cascaded baseline in Exp. 1 to the end-to-end VGGT-Seg (Vanilla) in Exp. 2 brings a substantial improvement from 15.1 to 21.8 mAP, validating the inherent advantages of eliminating explicit geometric dependencies. Building upon VGGT-Seg (Vanilla), introducing semantic priors significantly boosts the mAP by 8.0 to reach 29.8. This indicates that global semantic priors effectively break the limitations of geometric aliasing, guiding query points to anchor onto high-confidence foreground targets. Exp. 4 shows that relying solely on traditional FPS with hybrid query yields a smaller gain of 4.4 mAP, proving that without attention-guided sampling, queries lack precise initial localization. Furthermore, integrating only TQR (Exp. 5) or the complete TID (Exp. 6) into the vanilla model yields mAP improvements of 6.5 and 11.1, respectively. This demonstrates that explicit topological anchors and contrastive learning $\mathcal{L}_{icr}$ successfully align local receptive fields and repel spatially confusing negative samples, even without advanced query initialization. Individually removing components from the complete VGGT-Seg (Full) architecture (Exp. 7, 48.9 mAP) provides us with a completely new perspective. Removing A-FPS (Exp. 8) causes a significant performance drop of 6.2 mAP, underscoring that standard geometric sampling fails drastically in sparse and noisy regions; A-FPS is crucial for robust hybrid query generation. Omitting the instance-centric rectified contrastive loss $\mathcal{L}_{icr}$ (Exp. 9) results in a 2.3 mAP decrease, highlighting its role in ensuring boundary discrimination. Most notably, removing the entire TID module (Exp. 10) triggers a massive 10.5 mAP plummet (down to 38.4 mAP). This degradation proves that traditional decoding mechanisms suffer from spatial misalignments of supervisory signals on coarse point clouds, and TID is indispensable for stabilizing gradient descent. Moreover, adding SPQI (Exp. 3) or TID (Exp. 6) to the vanilla baseline individually yields 29.8 and 32.9 mAP, their combined integration in Exp. 7 pushes

TABLE V: Ablation study on replacing the frozen reconstruction backbone and isolating the impact of DINOv2-derived semantic priors.

Method	A-FPS	mAP	AP_{50}	AP_{25}
DUSt3R [21]	✗	39.5	58.1	69.3
MASt3R [22]	✗	40.3	59.7	71.5
π^3 [63]	✗	43.0	62.9	73.6
π^3 [63]	✓	48.2	**70.2**	81.2
VGGT [20]	✗	42.7	62.4	73.1
VGGT [20]	✓	**48.9**	70.1	**81.7**

TABLE VI: Ablation study on the impact of TID stacked layers L and query numbers $\{Q_{sem}, Q_{lrn}\}$.

Layers	Numbers	mAP	AP_{50}	AP_{25}
2	$\{100, 100\}$	43.7	66.0	75.8
4	$\{100, 100\}$	46.1	67.3	78.4
6	$\{100, 100\}$	47.2	68.8	79.1
8	$\{100, 100\}$	47.3	68.7	79.2
10	$\{100, 100\}$	OOM	OOM	OOM
6	$\{200, 200\}$	48.9	70.3	81.6
6	$\{300, 100\}$	49.1	71.2	82.2
6	$\{150, 150\}$	47.8	69.0	80.4
6	$\{200, 100\}$	48.9	70.1	81.7

the performance to 48.9 mAP. This demonstrates that SPQI and TID are highly complementary: SPQI provides explicitly anchored and semantically rich initial queries, which serve as a prerequisite for TID to effectively apply topological rectification. Conversely, without TID's progressive spatial constraints, the high-quality queries initialized by SPQI would degrade amidst the geometric noise during the decoding phase.

As illustrated in Fig. 5, we present qualitative comparisons to demonstrate the effectiveness of the SPQI module. Under the w/o SPQI setting, conventional geometric FPS struggles under sparse geometric degradation, where queries fail to achieve effective coverage. This leads to missing semantics, such as truncated chair leg structures in the first row. In the second row, adjacent cabinets are erroneously split into multiple instances, leading to instance misidentifications. In contrast, with the introduction of SPQI, the model leverages shallow semantic attention from DINOv2 to guide sampling and anchors onto the bodies of foreground targets via hybrid queries, effectively alleviating the issues of semantic omissions and instance misidentifications.

Impact of frozen reconstruction backbones and internal priors: To rigorously isolate the effect of the VGGT internal priors from our newly proposed segmentation modules, we conduct an ablation study replacing the frozen VGGT encoder with other equally strong state-of-the-art feed-forward 3D reconstruction backbones, as presented in Tab. V. Specifically, we evaluate DUSt3R [21], MASt3R [22], and π^3 [63]. Since DUSt3R and MASt3R do not natively incorporate DINOv2 in their architectures, they are evaluated using standard farthest point sampling (w/o A-FPS), yielding 39.5 and 40.3 **mAP**, respectively. For π^3 and VGGT, which both inherently utilize DINOv2, we evaluate them by removing the DINOv2-derived attention (w/o A-FPS) and then incorporating it (w/ A-FPS). Removing A-FPS leads to a noticeable performance drop for both π^3 (48.2 to 43.0 **mAP**) and VGGT (48.9 to 42.7 **mAP**). This confirms our segmentation architecture generalizes well across diverse foundation models, and A-FPS's semantic guidance robustly boosts performance.

TABLE VII: Performance and resource consumption across different numbers of input frames. We record the average inference time (s) and peak GPU memory (GB) consumption for each scene. OOM denotes for Out Of Memory.

Frames	Memory (GB)	Time (s)	mAP
10	5.7	1.8	42.5
20	8.7	2.5	43.8
40	11.2	5.0	47.4
60	15.4	7.2	48.0
80	23.8	10.4	48.9
100	OOM	OOM	OOM

TABLE VIII: Efficiency comparison of multi-view based methods on the ScanNet [60] validation set. We recorded the pre-processing time, total time, and peak GPU memory per scene using a single NVIDIA RTX 4090 GPU.

Method	Pre (s)	Total (s)	Memory (GB)
Mask3D† [9]	6.8	7.0	14.3
OneFormer3D† [7]	6.8	7.2	15.1
VGGT-Seg (Vanilla)	–	8.5	16.7
VGGT-Seg (Full)	–	10.4	23.8

Impact of TID stacked layers and query numbers: As shown in Tab. VI, while maintaining a fixed query configuration of $\{100, 100\}$, scaling the number of decoder layers from 2 to 6 yields a continuous improvement in **mAP** from 43.7 to 47.2. However, increasing the number of stacked layers to 8 brings only a marginal gain of 0.1 in overall metrics, indicating that the representational capacity of the model saturates rapidly after reaching a certain threshold. Furthermore, pushing the stacked layers to 10 triggers an out-of-memory (OOM). Therefore, the 6 layer configuration establishes the optimal balance between spatial perception accuracy and GPU memory efficiency. Subsequently, adjusting the query allocation from $\{100, 100\}$ to $\{200, 100\}$ leads to a substantial performance leap, reaching **mAP** of 48.9 and an $\mathbf{AP}_{50}$ of 70.1. Interestingly, further increasing the learnable queries to reach a $\{200, 200\}$ configuration does not significantly change the overall precision, suggesting that excessive implicit parameters introduce representational redundancy rather than discriminative capability. Although the highly dense $\{300, 100\}$ setup achieves 49.1, it suffers from severe diminishing marginal returns relative to its computational overhead. The $\{200, 100\}$ distribution emerges as the optimal choice for balancing robust instance

TABLE IX: Ablation study on the impact of different levels of noise. **w/o** and **w/** denote without and with the integration of the proposed **TQR**, respectively.

Noise Level	w/o TQR	w/ TQR
0.00	46.6	48.9
0.01	41.8	48.5
0.05	27.1	43.0
0.10	9.3	39.4
0.15	0.0	37.0
0.20	0.0	31.2

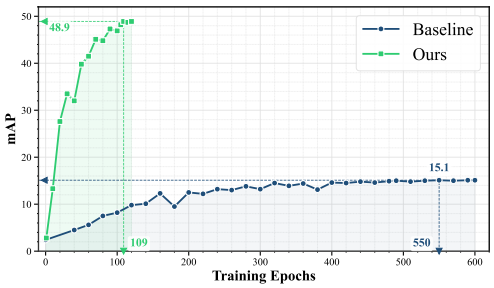


Fig. 6: Convergence and accuracy comparison of VGGT-Seg (Full) against baseline [10] under geometric degradation

localization and computational cost.

Impact of input frames and efficiency analysis: As shown in Tab. VII, as the number of input frames increases from 10 to 40, the segmentation accuracy improves significantly, with the **mAP** score jumping from 42.5 to 47.4. This indicates that the accumulation of multi-view observation data can significantly enrich the geometric and semantic priors required for scene understanding. Beyond the 40 frames, performance growth begins to saturate. Extending the input sequence from 40 to 80 frames brings a 1.5 improvement in **mAP**. Further expanding to 100 frames will lead to an out-of-memory error. This demonstrates that although dense view sequences can theoretically provide more comprehensive scene coverage, the heavy computational burden imposed by excessive frame inputs severely constrains the scalability.

Furthermore, we evaluate the inference efficiency of VGGT-Seg against other multi-view based methods, as shown in Tab. VIII. The Pre (s) column denotes the time consumed by the decoupled preprocessing stage, specifically the offline VGGT reconstruction process required by cascaded methods (e.g., Mask3D† and OneFormer3D†). VGGT-Seg adopts a fully end-to-end paradigm, directly inferring instance masks from sparse images without involving a preprocessing stage. Compared to VGGT-Seg (Vanilla), VGGT-Seg (Full) incurs higher total inference time (10.4s) and peak memory usage (23.8 GB) due to its joint semantic-geometric querying mechanism. However, given the substantial accuracy improvements it brings, VGGT-Seg (Full) achieves a significantly superior performance-efficiency trade-off over other methods.

Impact of different levels of noise: During the inference phase of VGGT-Seg, as shown in Tab. IX, we inject varying levels of Gaussian noise to simulate geometric degradation

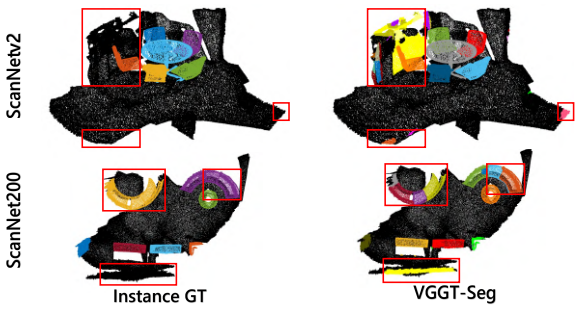


Fig. 7: Failure cases on the ScanNetv2 [60] and ScanNet200 [52] val dataset. Red boxes highlight over-segmentation on continuous structures.

in order to evaluate the robustness of the proposed TQR. As the noise level climbs from 0.00 to 0.05, **w/o TQR** experiences a performance decline, with the **mAP** plummeting from 46.6 to 27.1. **w/ TQR** effectively cushions this degradation. This significant performance gap demonstrates that explicitly constraining the spatial attention of queries using topological anchors successfully mitigates the model's sensitivity to noise, preserving its boundary discrimination capability even under moderate spatial corruption. As the noise intensity reaches 0.10 and 0.20, the **mAP** of **w/o TQR** drops to 0. This catastrophic failure stems from severe geometric aliasing, where unconstrained queries completely lose their localization targets amidst dense background noise. In contrast, **w/ TQR** retains effective predictive capabilities of 37.0 and 31.2, respectively. The robustness exhibited by the model under such dense background noise confirms that adaptive topological constraints fundamentally prevent queries from degrading into random spatial sampling, successfully yielding valid instance masks. Additionally, we recorded the mAP of the baseline and VGGT-Seg during the training phase, as shown in Fig. 6. Geometric degradation can lead to sampling bias and mismatched supervisory information, which is particularly evident in the baseline method, our method exhibits significantly higher accuracy and faster convergence speed.

However, our method still faces certain challenges, with representative failure cases illustrated in Fig. 7. Structural noise and depth projection bias tend to mislead semantic queries, resulting in over-segmentation. Specifically, unannotated black background regions in the ground truth are over-predicted as multiple independent instances, while continuous circular tables are over-segmented into multiple localized instance fragments. These failure cases provide promising directions for future research.

V. Conclusion

In this work, we presented VGGT-Seg, a novel end-to-end paradigm that liberates 3D instance segmentation from the strict dependencies on explicit sensor calibrations and dense depth maps. By treating sparse multi-view images as the sole input, a core advantage of our approach is its ability to systematically repurpose the internal priors of visual foundation models to tame the chaotic and sparse nature of feed-forward 3D reconstructions. By introducing Semantic Prior-

Driven Query Initialization, we successfully overcome the inherent localization biases of traditional geometric sampling, ensuring robust instance discovery even in highly degraded geometries. Furthermore, the Topology-rectified Interleaved Decoder introduces a dynamic mechanism to align supervisory signals and suppress background noise, effectively resolving the geometric aliasing that plagues sparse point clouds.

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